Year 8 Mathematics — Standard Worksheets Master Answer Key

Complete compiled answers for all 23 section printable worksheets (Sections 1A–10A · 230 questions / 506 parts, SymPy verified).

Chapter 1: Integers and Number Properties

1A · Adding and Subtracting Integers

VC2M8N04 · Standard Worksheet Answer Key

PART A · FLUENCY & SKILLS

Q1. **(a)** -5 **(b)** -7 **(c)** -42 **(d)** 15

Q2. **(a)** -17 **(b)** -31 **(c)** 42 **(d)** 8

Q3. **(a)** -6 **(b)** -6 **(c)** -16 **(d)** -12

Q4. **(a)** 23 **(b)** -17 **(c)** -19 **(d)** -16

PART B · PROBLEM-SOLVING & APPLICATIONS

Q5. (a) -9°C, (b) 5°C

Q6. $\$25$

Q7. $-88\ \text{m}$

Q8. $1986 - (-15) = 2001\ \text{m}$

PART C · REASONING & CHALLENGE

Q9. Always negative, because p lies to the left of q on the number line ($p < q \implies p - q < 0$; e.g. $-10 - (-3) = -7 < 0$).

Q10. $(-1, -8, 3)$

1B · Multiplying and Dividing Integers

VC2M8N04 · Standard Worksheet Answer Key

PART A · FLUENCY & SKILLS

Q1. **(a)** -56 **(b)** 108 **(c)** -90 **(d)** -120

Q2. **(a)** -8 **(b)** 8 **(c)** -12 **(d)** -9

Q3. **(a)** 36 **(b)** -36 **(c)** -64 **(d)** 144

Q4. **(a)** -24 **(b)** -12 **(c)** 15 **(d)** 8

PART B · PROBLEM-SOLVING & APPLICATIONS

Q5. (a) $-126\ \text{m}$, (b) $13\ \text{minutes}$

Q6. $-70\ \text{kL/day}$

Q7. $67\ \text{points}$

Q8. -3°C

PART C · REASONING & CHALLENGE

Q9. $(x, y) = (-8, 3)$, since $(-8) \times 3 = -24$ and $-8 + 3 = -5$.

Q10. Negative, because the total number of negative factors is $1 + 2 + 3 + 4 + 5 = 15$ (an odd number).

1C · Order of Operations and Directed Numbers

VC2M8N04 · Standard Worksheet Answer Key

PART A · FLUENCY & SKILLS

Q1. **(a)** -39 **(b)** 20 **(c)** 36 **(d)** 2

Q2. **(a)** -3 **(b)** -52 **(c)** 3 **(d)** -4

Q3. **(a)** 4 **(b)** -3 **(c)** -2 **(d)** -4

Q4. **(a)** 46 **(b)** -26 **(c)** -8 **(d)** -5

PART B · PROBLEM-SOLVING & APPLICATIONS

Q5. $\dfrac{3(-4)+5(-2)+2(3)}{10} = -1.6^\circ\text{C}$

Q6. $+\$260$ gain ($400(-3)+250(6)-40=260$)

Q7. -10°C

Q8. -9 m

PART C · REASONING & CHALLENGE

Q9. $-18 \div (3-9) + (-2)^3 = -18 \div (-6) + (-8) = 3 - 8 = -5$.

Q10. Sam subtracted $16-4$ before multiplying; Priya multiplied $4\times(-3)$ before squaring. Correct: $16-4(9)=16-36=-20$.

1D · LCM, HCF, and Prime Factorisation

VC2M8N02 · Standard Worksheet Answer Key

PART A · FLUENCY & SKILLS

Q1. **(a)** $2\times3^2\times5$ **(b)** $2^4\times3^2$ **(c)** $2^2\times3^2\times7$ **(d)** $2^2\times3^3\times5$

Q2. **(a)** 16 **(b)** 21 **(c)** 24 **(d)** 12

Q3. **(a)** 60 **(b)** 72 **(c)** 112 **(d)** 120

Q4. **(a)** $2^2\times3^2$ **(b)** 36 **(c)** $2^3\times3^3\times5\times7$ **(d)** 7560

PART B · PROBLEM-SOLVING & APPLICATIONS

Q5. 120 minutes (2 hours), at $9{:}00$ am

Q6. (a) 42 beds, (b) 2 native grasses and 3 kangaroo paws per bed

Q7. 5 rotations ($\text{LCM}(32,40)=160$ teeth; $160\div32=5$)

Q8. Side length $=24$ cm; 35 squares ($7\times5=35$)

PART C · REASONING & CHALLENGE

Q9. $k=7$ (so all prime exponents are even: $2^2\times3^2\times7^2=1764$), giving $\sqrt{1764}=42$.

Q10. 70, since $\dfrac{14\times210}{42}=70$.

Chapter 2: Real Numbers, Fractions, Decimals and Index Laws

2A · Operations with Positive and Negative Fractions

VC2M8N04 · Standard Worksheet Answer Key

PART A · FLUENCY & SKILLS

Q1. **(a)** $\frac{1}{8}$ **(b)** $-\frac{22}{15} = -1\frac{7}{15}$ **(c)** $\frac{29}{18} = 1\frac{11}{18}$ **(d)** $-\frac{7}{12}$

Q2. **(a)** $-\frac{1}{6}$ **(b)** $\frac{3}{8}$ **(c)** $-\frac{3}{2} = -1\frac{1}{2}$ **(d)** 3

Q3. **(a)** $-\frac{3}{4}$ **(b)** $\frac{10}{3} = 3\frac{1}{3}$ **(c)** -2 **(d)** $\frac{3}{2} = 1\frac{1}{2}$

Q4. **(a)** $-\frac{7}{6} = -1\frac{1}{6}$ **(b)** $-\frac{1}{3}$ **(c)** 1 **(d)** $-\frac{11}{4} = -2\frac{3}{4}$

PART B · PROBLEM-SOLVING & APPLICATIONS

Q5. $5\frac{2}{5}\ \text{km}$ ($\frac{27}{5}\ \text{km}$) **Q6.** $\frac{23}{30}$

Q7. 18 brackets **Q8.** -2°C

PART C · REASONING & CHALLENGE

Q9. $-\frac{13}{24}$, since $\frac{-\frac{10}{12} + (-\frac{3}{12})}{2} = -\frac{13}{24}$.

Q10. Dividing by $-\frac{2}{5}$ multiplies by the reciprocal $-\frac{5}{2}$, whose absolute value is $2.5 > 1$ (e.g. $6 \div (-\frac{2}{5}) = -15$).

2B · Terminating and Recurring Decimals

VC2M8N03 · Standard Worksheet Answer Key

PART A · FLUENCY & SKILLS

Q1. **(a)** Terminating ($40 = 2^3 \times 5$) **(b)** Recurring ($12 = 2^2 \times 3$) **(c)** Terminating ($\frac{3}{8} = \frac{3}{2^3}$) **(d)** Recurring ($30 = 2 \times 3 \times 5$)

Q2. **(a)** 0.6875 **(b)** $-0.8\dot{3}$ **(c)** $0.\dot{6}\dot{3}$ ($0.\overline{63}$) **(d)** $0.\dot{7}14285\dot{7}$

Q3. **(a)** $\frac{9}{20}$ **(b)** $-\frac{3}{8}$ **(c)** $\frac{41}{25} = 1\frac{16}{25}$ **(d)** $\frac{1}{16}$

Q4. **(a)** $\frac{7}{9}$ **(b)** $\frac{5}{11}$ **(c)** $\frac{5}{37}$ **(d)** $\frac{5}{18}$

PART B · PROBLEM-SOLVING & APPLICATIONS

Q5. (a) $\frac{5}{12}$, (b) $0.41\dot{6}$ **Q6.** $\frac{82}{99}\ \text{g}$ ($\frac{44+27+11}{99} = \frac{82}{99}\ \text{g}$)

Q7. $0.6 < 0.\dot{6}\dot{0} < 0.66 < 0.\dot{6}$ **Q8.** 0.34375 inches

PART C · REASONING & CHALLENGE

Q9. You must simplify the fraction BEFORE testing the denominator: $\frac{18}{150} = \frac{3}{25} = \frac{3}{5^2} = 0.12$, which terminates.

Q10. $10x - x = 9.\dot{9} - 0.\dot{9} \implies 9x = 9 \implies x = 1$; also $0.\dot{3} + 0.\dot{6} = 0.\dot{9} = \frac{1}{3} + \frac{2}{3} = 1$.

2C · Irrational Numbers and the Real Number System

VC2M8N01 · Standard Worksheet Answer Key

PART A · FLUENCY & SKILLS

Q1. **(a)** Rational ($= 11$) **(b)** Irrational (40 is not a perfect square) **(c)** Rational ($= -\frac{9}{11}$) **(d)** Irrational (non-zero rational multiple of π)

Q2. **(a)** 13 **(b)** $\dfrac{7}{12}$ **(c)** -5 **(d)** 0.6

Q3. **(a)** Between 5 and 6 ($25 < 31 < 36$) **(b)** Between 9 and 10 ($81 < 88 < 100$) **(c)** Between -8 and -7 **(d)** Between 3 and 4 ($27 < 50 < 64$)

Q4. **(a)** 16 **(b)** 28 **(c)** 15 **(d)** 10

PART B · PROBLEM-SOLVING & APPLICATIONS

Q5. (a) $\sqrt{75}$ m, (b) between 8 m and 9 m ($64 < 75 < 81$)

Q6. $C = 22\pi$ m, $A = 121\pi$ m^2; both are irrational.

Q7. $\dfrac{22}{7} \approx 3.1429 > \pi \approx 3.1416$, so $\dfrac{22}{7}$ is slightly larger.

Q8. 70 cm ($\sqrt[3]{343\,000} = 70$ cm)

PART C · REASONING & CHALLENGE

Q9. Product: $\sqrt{2} \times \sqrt{8} = \sqrt{16} = 4$ (or $\sqrt{3} \times \sqrt{3} = 3$). Sum: $\pi + (4 - \pi) = 4$ (or $\sqrt{5} + (-\sqrt{5}) = 0$).

Q10. The square of any real number is non-negative ($x^2 \geq 0$), so no real number squares to -36; however, an odd power of a negative number is negative, so $(-4)^3 = -64$.

2D · Index Laws and the Zero Exponent

VC2M8N02 · Standard Worksheet Answer Key

PART A · FLUENCY & SKILLS

Q1. **(a)** 3^9 **(b)** 7^9 **(c)** x^{11} **(d)** $20a^8$

Q2. **(a)** 5^4 **(b)** 2^6 **(c)** m^6 **(d)** $4x^5$

Q3. **(a)** 2^{12} **(b)** m^8 **(c)** 8 **(d)** 0

Q4. **(a)** $3^3 = 27$ **(b)** $2^5 = 32$ **(c)** $5^2 = 25$ **(d)** $2^3 = 8$

PART B · PROBLEM-SOLVING & APPLICATIONS

Q5. $2^{11} = 2048$ files

Q6. $3^2 \times 3^5 = 3^7 = 2187$ cells

Q7. (a) $(2^3)^2 = 2^6 = 64$ cm^2, (b) $(2^3)^3 = 2^9 = 512$ cm^3

Q8. $n = 5$ ($2n + 4 - 5 = 9 \implies 2n = 10 \implies n = 5$)

PART C · REASONING & CHALLENGE

Q9. Index laws require the SAME base. Since $4 = 2^2$, $4^3 = (2^2)^3 = 2^6$, so $4^3 \times 2^5 = 2^6 \times 2^5 = 2^{11} = 2048$.

Q10. 3^{20} is larger, because $2^{30} = (2^3)^{10} = 8^{10}$ whereas $3^{20} = (3^2)^{10} = 9^{10}$, and $9^{10} > 8^{10}$.

Chapter 3: Percentages and Financial Mathematics

3A · Percentages, Percentage Change and GST

VC2M8N05 · Standard Worksheet Answer Key

PART A · FLUENCY & SKILLS

Q1. **(a)** $\$36$ **(b)** $63\ \text{kg}$ **(c)** $\$27$ **(d)** $80\ \text{L}$

Q2. **(a)** $\$92$ **(b)** $\$114$ **(c)** $486\ \text{m}$ **(d)** $\$280$

Q3. **(a)** 24% increase **(b)** 15% decrease **(c)** 28% increase **(d)** 22% decrease

Q4. **(a)** $\$14$ **(b)** $\$286$ **(c)** $\$180$ **(d)** $\$42$

PART B · PROBLEM-SOLVING & APPLICATIONS

Q5. (a) $\$48$, (b) $\$528$

Q6. (a) $\$290$, (b) $\$29$

Q7. 18% increase

Q8. $\$240$ ($192 \div 0.80 = 240$)

PART C · REASONING & CHALLENGE

Q9. $\$400 \times 1.25 = \$500$, and $\$500 \times 0.80 = \$400$. Since $\frac{5}{4} \times \frac{4}{5} = 1$, a 20% reduction of the larger amount equals 25% of the original amount.

Q10. The 10% GST was calculated on the pre-GST price (10% of $\$200 = \$20$), whereas 10% of $\$220$ is $\$22$. To undo $\times 1.1$, you must divide by 1.1.

3B · Profit, Loss, Discounts and Simple Interest

VC2M8N05 · Standard Worksheet Answer Key

PART A · FLUENCY & SKILLS

Q1. **(a)** $\$15$ profit (25% profit) **(b)** $\$24$ loss (20% loss) **(c)** $\$36$ profit (45% profit) **(d)** $\$35$ loss (14% loss)

Q2. **(a)** $\$117$ **(b)** $\$136$ **(c)** $\$324$ **(d)** $\$252$

Q3. **(a)** $\$300$ **(b)** $\$900$ **(c)** $\$720$ **(d)** $\$108$

Q4. **(a)** $\$5720$ **(b)** $4\ \text{years}$ **(c)** $4.5\%\ \text{p.a.}$ **(d)** $\$2500$

PART B · PROBLEM-SOLVING & APPLICATIONS

Q5. (a) $\$240$, (b) $\$180$, (c) 20% profit ($\$30$ profit on $\$150$)

Q6. (a) $\$1014$, (b) $\$7514$

Q7. Commission $= \$16\,500$; vendor receives $\$733\,500$

Q8. $\$864$; equivalent to a 28% discount

PART C · REASONING & CHALLENGE

Q9. $12.5\ \text{years}$, since $I = Prt = P \implies rt = 1 \implies 0.08t = 1 \implies t = \frac{1}{0.08} = 12.5\ \text{years}$.

Q10. Store A multiplies by $0.70 \times 0.90 = 0.63$; Store B multiplies by $0.90 \times 0.70 = 0.63$ (commutative property of multiplication), so both give a 37% total discount.

Chapter 4: Ratios, Rates and Best Buys

4A · Simplifying Ratios and Dividing in a Ratio

VC2M8M07 · Standard Worksheet Answer Key

PART A · FLUENCY & SKILLS

Q1. **(a)** $2:3$ **(b)** $7:3$ **(c)** $5:8$ **(d)** $2:3:5$

Q2. **(a)** $3:4$ **(b)** $9:10$ **(c)** $1:3$ **(d)** $3:8$

Q3. **(a)** $\$60$ and $\$80$ **(b)** $60\ \text{kg}$ and $36\ \text{kg}$ **(c)** $100\ \text{mL}$ and $150\ \text{mL}$ **(d)** $\$80$, $\$120$, and $\$160$

Q4. **(a)** $x = 49$ **(b)** $x = 35$ **(c)** $30\ \text{m}$ **(d)** $9\ \text{km}$

PART B · PROBLEM-SOLVING & APPLICATIONS

Q5. (a) $60\ \text{kg}$, (b) $120\ \text{kg}$, (c) $240\ \text{kg}$

Q6. Zara invested $\$63\,000$; combined investment $= \$108\,000$

Q7. $3.8\ \text{km}$ ($380\,000\ \text{cm} = 3800\ \text{m} = 3.8\ \text{km}$)

Q8. $40°, 60°, 80°$; acute-angled triangle

PART C · REASONING & CHALLENGE

Q9. $6:4:5$ (multiply $3:2$ by 2 to match pineapple $= 4$ parts).

Q10. 16 red marbles (currently 24 red and 40 blue; $40 - 24 = 16$).

4B · Rates, Speed and Best Buys — VC2M8M05 · Standard Worksheet Answer Key

PART A · FLUENCY & SKILLS

Q1. **(a)** $80\ \text{km/h}$ **(b)** $\$23/\text{h}$ **(c)** $12.5\ \text{km/L}$ **(d)** $62.5\ \text{words/min}$

Q2. **(a)** $240\ \text{km}$ **(b)** $3.5\ \text{hours}$ (3 h 30 min) **(c)** $8\ \text{m/s}$ **(d)** $260\ \text{m}$

Q3. **(a)** $54\ \text{km/h}$ **(b)** $30\ \text{m/s}$ **(c)** $90\ \text{km/h}$ **(d)** $0.5\ \text{L/s}$

Q4. **(a)** $\$1.80$ per $100\ \text{g}$ **(b)** $\$0.65$ per $100\ \text{g}$ **(c)** $\$12.60/\text{L}$ **(d)** $\$1.55$ per $100\ \text{g}$

PART B · PROBLEM-SOLVING & APPLICATIONS

Q5. Pack A $= \$2.60/100\ \text{g}$, Pack B $= \$2.30/100\ \text{g}$, Pack C $= \$2.40/100\ \text{g}$; Pack B is the best buy.

Q6. (a) $30.6\ \text{L}$, (b) $\$59.67$

Q7. $90\ \text{km/h} = 25\ \text{m/s}$

Q8. $30\ \text{minutes}$ (combined rate $= 20\ \text{L/min}$; $600 \div 20 = 30$)

PART C · REASONING & CHALLENGE

Q9. Total distance $= 60\ \text{km}$ and total time $= 2 + 1 = 3\ \text{hours}$, so $\text{Average speed} = \dfrac{60}{3} = 20\ \text{km/h}$.

Q10. In $1\ \text{hour}$ ($3600\ \text{s}$), an object moving at $1\ \text{m/s}$ travels $3600\ \text{m} = \dfrac{3600}{1000}\ \text{km} = 3.6\ \text{km}$, so $1\ \text{m/s} = 3.6\ \text{km/h}$.

Chapter 5: Algebraic Techniques

5A · Language of Algebra, Substitution and Like Terms — VC2M8A01 · Standard Worksheet Answer Key

PART A · FLUENCY & SKILLS

Q1. **(a)** 29 **(b)** 17 **(c)** 98 **(d)** -8

Q2. **(a)** $-3a + 4b$ **(b)** $4x^2 - 9x - 3$ **(c)** $9mn - 9m^2n$ **(d)** $-3pq^2 - 2p^2q$

Q3. **(a)** $-30ab$ **(b)** $28x^5$ **(c)** $-24m^3n^3$ **(d)** $-24a^4b^4$

Q4. **(a)** $-4x^3$ **(b)** $3a^2b$ **(c)** $-\dfrac{2m}{3n}$ **(d)** $-3y^2$

PART B · PROBLEM-SOLVING & APPLICATIONS

Q5. (a) $P = (14x + 2)\ \text{m}$, (b) $44\ \text{m}$

Q6. $12g + 3b + 17\ \text{points}$

Q7. $(24ab - 4a^2)\ \text{cm}^2$

Q8. $70h + 15v + 85$; $\$355$

PART C · REASONING & CHALLENGE

Q9. By the commutative law, $yx^2 = x^2y$, so both have x^2 and y^1; in xy^2, x has power 1 and y has power 2 ($2^2(3) = 12 \neq 2(3^2) = 18$).

Q10. $n + (n+1) + (n+2) + (n+3) = 4n + 6 = 2(2n + 3)$, which is always even and leaves a remainder of 2 when divided by 4.

5B · Expanding Brackets and Factorising Linear Expressions — VC2M8A01 · Standard Worksheet Answer Key

PART A · FLUENCY & SKILLS

Q1. **(a)** $10x - 35$ **(b)** $-12a - 20$ **(c)** $-24 + 18m = 18m - 24$ **(d)** $12x^2 - 15xy$

Q2. **(a)** $22x - 7$ **(b)** $3a - 18$ **(c)** $8m - 5$ **(d)** $x^2 - x$

Q3. **(a)** $5(3x + 5)$ **(b)** $6(3a - 4b)$ **(c)** $7m(2m - 3)$ **(d)** $8xy(2x + 3y)$

Q4. **(a)** $-4(2x + 5)$ **(b)** $-6(3a - 2)$ **(c)** $-5m(3m + 5)$ **(d)** $-7pq(3p - 2)$

PART B · PROBLEM-SOLVING & APPLICATIONS

Q5. (a) $8(3x + 5)\ \text{m}^2$, (b) $(24x + 40)\ \text{m}^2$

Q6. $10x + 36$

Q7. Width $= 7x\ \text{cm}$, length $= (4x - 5)\ \text{cm}$

Q8. Expanded: $(16x + 5)\ \text{cm}$ (cannot be factorised further over integers)

PART C · REASONING & CHALLENGE

Q9. Multiplication $-3(2x - 5)$ must be performed before subtracting from 14. Correct: $14 - 6x + 15 = 29 - 6x$.

Q10. $10n + 15 - 6n + 3 + 2n - 6 = 6n + 12 = 6(n + 2)$, which has a factor of 6 for every integer n.

Chapter 6: Linear Equations and Coordinate Graphs

6A · Solving Linear Equations — VC2M8A02 · Standard Worksheet Answer Key

PART A · FLUENCY & SKILLS

Q1. **(a)** $x = 7$ **(b)** $x = -3$ **(c)** $x = -5$ **(d)** $x = 42$

Q2. **(a)** $x = 11$ **(b)** $x = -12$ **(c)** $x = 5$ **(d)** $x = -3$

Q3. **(a)** $x = 7$ **(b)** $x = -6$ **(c)** $x = 6$ **(d)** $x = 5$

Q4. **(a)** $x = 4$ **(b)** $x = 6$ **(c)** $x = 8$ **(d)** $x = 6.5\ (\frac{13}{2})$

PART B · PROBLEM-SOLVING & APPLICATIONS

Q5. Width $w = 11\ \text{m}$, length $l = 25\ \text{m}$

Q6. $3\ \text{hours}$ (at $\$96$)

Q7. $33, 34, 35, 36$

Q8. Child ticket $= \$26$, adult ticket $= \$40$

PART C · REASONING & CHALLENGE

Q9. $m = 2$, since $3(8 - m) = 18 \implies 8 - m = 6 \implies m = 2$.

Q10. Infinitely many solutions, because expanding both sides gives the identity $4x - 12 = 4x - 12$, which is true for every real number x.

6B · Graphing Linear Relations and Modelling — VC2M8A03 · Standard Worksheet Answer Key

PART A · FLUENCY & SKILLS

Q1. **(a)** $m = 5, c = -8$ **(b)** $m = -3, c = 11$ **(c)** $m = -2, c = 9$ **(d)** $m = \frac{3}{4}, c = 0$

Q2. **(a)** $m = 3$ **(b)** $m = -3$ **(c)** $m = 4$ **(d)** $m = -0.5$ $(-\frac{1}{2})$

Q3. **(a)** $(0, -12)$ and $(4, 0)$ **(b)** $(0, 10)$ and $(5, 0)$ **(c)** $(0, 4)$ and $(5, 0)$ **(d)** $(0, -9)$ and $(6, 0)$

Q4. **(a)** $y = 4x - 7$ **(b)** $y = 4x + 5$ **(c)** $y = -2x - 3$ **(d)** $y = 3x + 1$

PART B · PROBLEM-SOLVING & APPLICATIONS

Q5. (a) $T = 8t + 20$, (b) 10 minutes

Q6. (a) $D = -1.5t + 45$, (b) 18 GB, (c) 30 days

Q7. $d = -75t + 150$; d-intercept $(0, 150)$ is starting distance (150 km); t-intercept $(2, 0)$ is arrival time (2 hours).

Q8. Intercepts $(9, 0)$ and $(0, 6)$; Area $= \frac{1}{2} \times 9 \times 6 = 27$ square units

PART C · REASONING & CHALLENGE

Q9. Both lines have the exact same gradient ($m = 3$), so they rise at the same rate, but they have different y-intercepts ($-4 \neq 5$), so they stay 9 vertical units apart everywhere.

Q10. Yes, they are collinear because $m_{PQ} = \frac{4 - (-5)}{2 - (-1)} = 3$ and $m_{QR} = \frac{16 - 4}{6 - 2} = 3$.

Chapter 7: Length, Area, Volume and Time

7A · Perimeter and Area of Quadrilaterals and Composite Shapes — VC2M8M01 · Standard Worksheet Answer Key

PART A · FLUENCY & SKILLS

Q1. **(a)** 720 mm^2 **(b)** $35\,000$ cm^2 **(c)** 4.8 ha **(d)** 160 ha

Q2. **(a)** 144 cm^2 **(b)** 100 m^2 **(c)** 126 cm^2 **(d)** 60 m^2

Q3. **(a)** 66 cm^2 **(b)** 180 m^2 **(c)** 80 cm^2 **(d)** 180 m^2

Q4. **(a)** 9 cm **(b)** 14 cm **(c)** 10 m **(d)** 16 cm

PART B · PROBLEM-SOLVING & APPLICATIONS

Q5. (a) 48 m, (b) 116 m^2

Q6. Outer dimensions 50 cm $\times$ 40 cm; frame area $= 800$ cm^2

Q7. $300\,000$ m$^2 = 30$ ha

Q8. $\$63$ (Area $= 1.8$ m^2; $1.8 \times 35 = 63$)

PART C · REASONING & CHALLENGE

Q9. The triangle's height must be twice the parallelogram's height ($h_{\text{tri}} = 2h_{\text{para}}$), since $\frac{1}{2}bh_{\text{tri}} = bh_{\text{para}} \implies h_{\text{tri}} = 2h_{\text{para}}$.

Q10. The horizontal and vertical inner step edges equal the removed top and side edges in length, so the total boundary length stays $2(W + H)$.

7B · Circumference and Area of Circles and Sectors

VC2M8M02 · Standard Worksheet Answer Key

PART A · FLUENCY & SKILLS

Q1. **(a)** 14π cm **(b)** 18π m **(c)** 9π cm **(d)** 25π mm

Q2. **(a)** 36π cm^2 **(b)** 121π m^2 **(c)** 64π cm^2 **(d)** 225π m^2

Q3. **(a)** $A = 50\pi$ cm^2, $P = (10\pi + 20)$ cm **(b)** $A = 9\pi$ m^2, $P = (3\pi + 12)$ m **(c)** $A = \dfrac{27\pi}{2}$ cm^2, $P = (3\pi + 18)$ cm **(d)** $A = 48\pi$ cm^2, $P = (8\pi + 24)$ cm

Q4. **(a)** 132 cm **(b)** 154 m^2 **(c)** 36 cm **(d)** 154 cm^2

PART B · PROBLEM-SOLVING & APPLICATIONS

Q5. (a) 440 m, (b) $15\,400$ m$^2 = 1.54$ ha

Q6. 20π m$^2 \approx 62.83$ m^2

Q7. 675π cm^2

Q8. $(200 - 50\pi)$ cm$^2 \approx 42.92$ cm^2

PART C · REASONING & CHALLENGE

Q9. Square side $= 11$ cm $\implies A_{\text{square}} = 121$ cm^2; Circle radius $= 7$ cm $\implies A_{\text{circle}} = 154$ cm^2. The circle encloses 33 cm^2 more area.

Q10. $2\pi r = 16\pi \implies r = 8$ cm, so $A = \pi(8)^2 = 64\pi$ cm^2.

7C · Volume, Capacity of Prisms and Time Zones

VC2M8M03 · Standard Worksheet Answer Key

PART A · FLUENCY & SKILLS

Q1. **(a)** 650 mL **(b)** 14.5 L **(c)** 3800 L **(d)** 25 m^3

Q2. **(a)** 480 cm^3 **(b)** 490 cm^3 **(c)** 560 cm^3 **(d)** 180 m^3

Q3. **(a)** $15{:}45$ **(b)** $12{:}25$ am **(c)** 4 h 40 min **(d)** $2{:}15$ pm ($14{:}15$)

Q4. **(a)** $2{:}00$ pm **(b)** $1{:}30$ pm **(c)** $12{:}00$ pm (noon) **(d)** $4{:}00$ pm

PART B · PROBLEM-SOLVING & APPLICATIONS

Q5. (a) 63 m^3, (b) $63\,000$ L

Q6. 324 cm^3

Q7. $8{:}55$ am AEST ($9{:}55$ am AEDT minus 1 hour)

Q8. 1.5 m ($4.0h = 6 \implies h = 1.5$ m)

PART C · REASONING & CHALLENGE

Q9. 4 cm, since the cube displaces $10^3 = 1000$ cm^3 of water, and $\Delta h = \dfrac{1000}{250} = 4$ cm.

Q10. In winter Melbourne is 2 hours ahead of Perth ($10 - 8 = 2$), whereas in December Melbourne observes Daylight Saving (UTC $+11$) while WA does not, making the difference $11 - 8 = 3$ hours.

Chapter 8: Pythagoras' Theorem and Geometry

8A · Pythagoras' Theorem

VC2M8M06 · Standard Worksheet Answer Key

PART A · FLUENCY & SKILLS

Q1. **(a)** $10\ \text{cm}$ **(b)** $13\ \text{m}$ **(c)** $25\ \text{cm}$ **(d)** $41\ \text{m}$

Q2. **(a)** $6\ \text{cm}$ **(b)** $24\ \text{m}$ **(c)** $16\ \text{cm}$ **(d)** $20\ \text{m}$

Q3. **(a)** $\sqrt{65}\ \text{cm} \approx 8.06\ \text{cm}$ **(b)** $\sqrt{117}\ \text{m} \approx 10.82\ \text{m}$ **(c)** $\sqrt{85}\ \text{cm} \approx 9.22\ \text{cm}$ **(d)** $\sqrt{115}\ \text{m} \approx 10.72\ \text{m}$

Q4. **(a)** Yes ($81 + 144 = 225 = 15^2$) **(b)** No ($49 + 100 = 149 \neq 144$) **(c)** Yes ($144 + 256 = 400 = 20^2$) **(d)** Yes ($2.25 + 4.0 = 6.25 = 2.5^2$)

PART B · PROBLEM-SOLVING & APPLICATIONS

Q5. $\sqrt{15\,649}\ \text{m} \approx 125.10\ \text{m}$

Q6. $4.8\ \text{m}$

Q7. $20\ \text{km}$

Q8. $h = \sqrt{75}\ \text{cm} \approx 8.66\ \text{cm}$

PART C · REASONING & CHALLENGE

Q9. $2s^2 = 144 \implies s^2 = 72$. Since the area of a square of side s is s^2, the exact area is immediately $72\ \text{cm}^2$.

Q10. $(ka)^2 + (kb)^2 = k^2a^2 + k^2b^2 = k^2(a^2 + b^2) = k^2c^2 = (kc)^2$.

8B · Congruence, Similarity and Quadrilateral Properties

VC2M8SP01 · Standard Worksheet Answer Key

PART A · FLUENCY & SKILLS

Q1. **(a)** SSS **(b)** SAS **(c)** ASA **(d)** RHS

Q2. **(a)** $x = 67$ **(b)** $x = 116$ **(c)** $x = 107$ **(d)** $x = 109$

Q3. **(a)** $k = 2.5$ **(b)** $15\ \text{cm}$ **(c)** $17.5\ \text{cm}$ **(d)** $5 : 2$ ($42.5 : 17 = 2.5 : 1$)

Q4. **(a)** Rhombus **(b)** Rectangle **(c)** Kite **(d)** Trapezium

PART B · PROBLEM-SOLVING & APPLICATIONS

Q5. $23\ \text{m}$

Q6. (a) $BD = 17\ \text{cm}$, (b) $BC = 15\ \text{cm}$ ($\sqrt{17^2 - 8^2} = 15$)

Q7. SSS (or RHS since $AM \perp BC$); $AM = 2.5\ \text{m}$

Q8. $x = 45$; angles are $60^\circ, 85^\circ, 100^\circ, 115^\circ$

PART C · REASONING & CHALLENGE

Q9. Equal angles fix the shape and side ratios of a triangle, but one triangle can be an enlargement of the other with side lengths multiplied by any scale factor $k \neq 1$.

Q10. In $\triangle ABC$ and $\triangle BAD$: $BC = AD$ (opposite sides of rectangle), $\angle ABC = \angle BAD = 90^\circ$, and $AB = BA$ (common side). Hence $\triangle ABC \cong \triangle BAD$ (SAS), so hypotenuse $AC = BD$.

Chapter 9: Probability and Venn Diagrams

9A · Complementary Events, Venn Diagrams and Two-Way Tables — VC2M8P02 · Standard Worksheet Answer Key

PART A · FLUENCY & SKILLS

Q1. **(a)** 0.73 **(b)** $\frac{7}{12}$ **(c)** 36% (0.36) **(d)** $\frac{6}{25}$ (0.24)

Q2. **(a)** 16 **(b)** 10 **(c)** 38 **(d)** $\frac{6}{25} = 0.24$ $(\frac{12}{50})$

Q3. **(a)** 0.80 **(b)** 0.30 **(c)** 0.20 **(d)** 0.20

Q4. **(a)** $\frac{3}{10} = 0.3$ **(b)** $\frac{1}{2} = 0.5$ (5 vowels) **(c)** $\frac{7}{10} = 0.7$ **(d)** $\frac{2}{5} = 0.4$ $(\mathrm{S}, \mathrm{T}, \mathrm{R}, \mathrm{L})$

PART B · PROBLEM-SOLVING & APPLICATIONS

Q5. (a) 15, (b) $\frac{23}{60}$

Q6. $n(W \cap B') = 17, n(W' \cap B) = 44, n(W' \cap B') = 31$; $\Pr(W' \cap B') = \frac{31}{120}$

Q7. (a) $\frac{3}{16}$, (b) $\frac{7}{16}$ $(1 - \frac{9}{16} = \frac{7}{16})$

Q8. $\frac{1}{3}$ $(\frac{6}{10} \times \frac{5}{9} = \frac{30}{90} = \frac{1}{3})$

PART C · REASONING & CHALLENGE

Q9. Maximum $= 0.45 + 0.35 = 0.80$ (when $\Pr(A \cap B) = 0$); Minimum $= 0.45$ (when $B \subseteq A$ so $\Pr(A \cap B) = 0.35$).

Q10. Only 2 of the 8 outcomes fail to have both a Head and a Tail (HHH and TTT), leaving $8 - 2 = 6$ favourable outcomes: $\frac{6}{8} = \frac{3}{4}$.

Chapter 10: Statistics and Data Analysis

10A · Data Collection, Summary Statistics and Outliers — VC2M8ST02 · Standard Worksheet Answer Key

PART A · FLUENCY & SKILLS

Q1. **(a)** 14 **(b)** 13 **(c)** 16 **(d)** 17

Q2. **(a)** 15 **(b)** 15 **(c)** 20 **(d)** 20

Q3. **(a)** 2 **(b)** 4 **(c)** 2 **(d)** 2.55

Q4. **(a)** Census **(b)** Sample **(c)** Primary data **(d)** Secondary data

PART B · PROBLEM-SOLVING & APPLICATIONS

Q5. (a) Median $= \$540\text{k}$, Mean $= \$665.71\text{k}$; (b) New Median $= \$540\text{k}$ (unchanged!), New Mean $= \$543.33\text{k}$

Q6. 49 goals

Q7. 79

Q8. Outlier $= 65$ min; Median $= 24$ min, Mean $= 26.56$ min $(\frac{239}{9})$, Range $= 47$ min

PART C · REASONING & CHALLENGE

Q9. 25, since $6 \times 15 - 5 \times 13 = 90 - 65 = 25$.

Q10. $5, 5, 9, 13$ ($a = b = 5$ for mode 5; $\frac{5+c}{2} = 7 \implies c = 9$; $5 + 5 + 9 + d = 32 \implies d = 13$).